The Knowledge Academy

Lean 6 Sigma Yellow & Green Belt

Introduction to the course

Contents:

Your training starts here!

Lean 6 Sigma primer (White Belt)

Course Case Study

Your training starts here!

Many thanks for choosing to study with The Knowledge Academy. We are sure that you will recognise the need to prepare for any course of study and have thus created these notes to get you started on your journey into Lean 6 Sigma

This guide has been written to help you understand the teaching framework which we as a company offer so that you arrive on you first day relaxed and prepared.

In particular we will cover:

- The preparation you will need to perform before the first day
- What is provided for you
- What you should bring
- A brief introduction to the subject (We call this “White Belt Training”)
- And a brief introduction to the exams

IMPORTANT POINT *Please make sure you complete all the pre-course work. The course is intensive and you will need an understanding of the topics mentioned here to allow you to gain the most from your investment*

What’s in this workbook?

This Workbook forms part of a set of material which you will need to efficiently prepare for the course. All of the documents should have been attached to the e-mail that included this file. This workbook consists of three sections

1. Guidance on how to prepare for the course
2. An introduction to Lean 6 Sigma
3. The Case Study scenario used on the course (both for Yellow & Green Belt segments)

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Important information on how to prepare for the course

- We suggest that you bring everything you need to study with you on the first day, this will typically include:
  - ◊ Pens & Pencils
  - ◊ Highlighters
  - ◊ Page Tabs
  - ◊ Note paper
  - ◊ Any other stationary

  Additionally, if you are booked on the Green Belt course it is advisable to bring the following:
  - ◊ A simple calculator (many people have this functionality on their phones)
  - ◊ Ruler
  - ◊ Graph paper

- Arrive on time: We start promptly and the intensive nature of the course means that if you miss something you may not have the time to catch up; also if you come in late you will distract other students. Please make sure you are in the class on time both at the start of the day and returning after breaks. Our courses start at 09:00 promptly on the first day, please be on the premises by 08:45 so that you have time to get yourself settled in.

- Unless you have been specifically informed otherwise we do not provide lunch at our venues (we do provide tea, coffee and chilled water) however there are always food and drink outlets close by. You may wish to bring in a cold lunch rather than purchase on the day.

- Additional Support: If you feel that you need additional support on the course, or with the exams, please contact your Sales Advisor **Before** the course so that your needs can be assessed.

Pre-course preparation

We suggest that the following short preparation will optimise your time on the course

1. Study the “L6S Primer ” presentation (on the next page)
2. Review the Case Study (at the end of this document). You do not need to memorise this but do make yourself aware of the contents as the Case Study is used in Class Exercises and will be referenced in the Green Belt Exam
3. Plan your week to minimise outside distractions and make sure you have everything you need ready for the start of the course

Thank you

The Lean 6 Sigma training team

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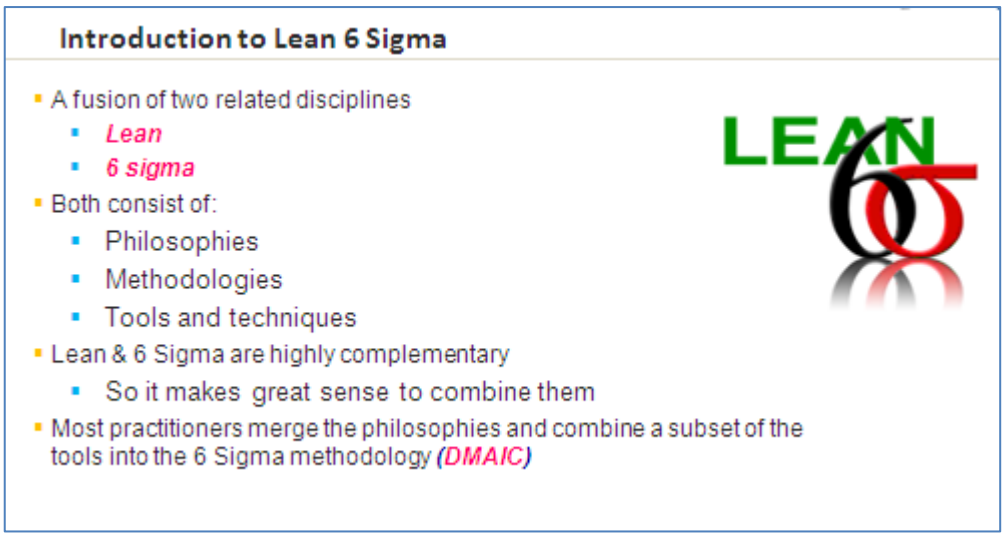
Lean 6 Sigma Primer

Introduction

The aim of this primer is to give you a basic overview of what Lean 6 Sigma is, what the belts do and the structure of the course. We have used a combination of graphic slides and text to minimise your study time.

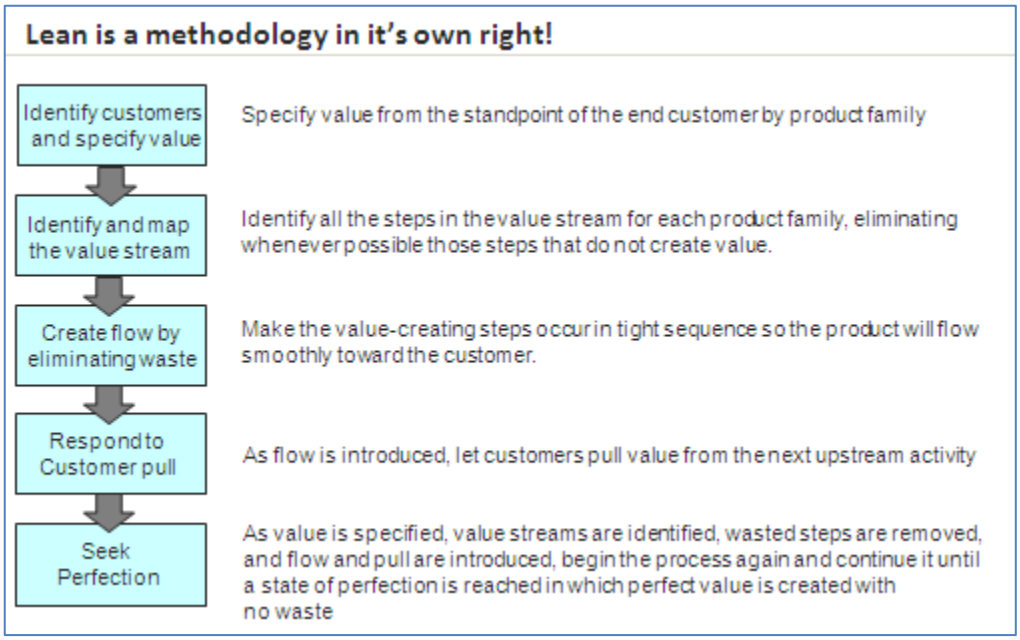
Please work through this material rapidly but thoroughly, researching any topics that you have difficulty with.



L6S (Lean 6 Sigma) is a combination of two separate quality improvement systems that arose out of Manufacturing. Lean grew out of **TPS** (Toyota Production System) while 6 Sigma came from Motorola (a big electronic components manufacturer). There is a remarkable synergy between the two systems such that using them together is really a case of the sum being better than the parts.

Lean & 6 Sigma are not really standardised in the real world. Many people have different interpretations of just what constitutes Lean, 6 Sigma and L6S. When writing the course we have had to make decisions about what to include and how to describe it. If you research what we teach you will often find several different explanations so be careful on the Internet. In our courses we have attempted to give you a unified story when in fact no such thing exists!

Overview of Lean



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People will always tell you that there is a methodology to Lean (an order to do things), however Lean is often considered to be quite unstructured as the techniques often do not need a framework to deploy them. By this I mean you can use the techniques we discuss on the course in any order, combination and at any time (at home or at work for that matter). This slide is in our presentation to give you the "rounded picture", the most common practice in L6S is to adopt the 6 Sigma methodology (which we will look at later) and add in the Lean tools as required.

About Lean

- The Lean methodology focuses on eliminating waste and smoothing the process flow.
  - Waste is anything perceived by the customer as not adding value to the products or services they consume.
- The groundwork for lean is the Toyota Production System (TPS).
- TPS influenced many other Japanese manufacturers and came to world-wide attention in the book **"The Machine that Changed the World"**
- Lean is **not** about cutting costs it is about removing waste without sacrificing quality

Typical Lean Techniques:

- Waste
- VOC (Voice Of Customer)
- Kaizen
- Single Piece Flow/ Pull Production/ Process Levelling
- Gemba
- Total staff involvement

We will introduce the concept of "waste" next, the other tools will be described during the classroom course however you might want to do a bit of pre-study if you have the time.

Lean focuses on increasing value and reducing inefficiencies

Value

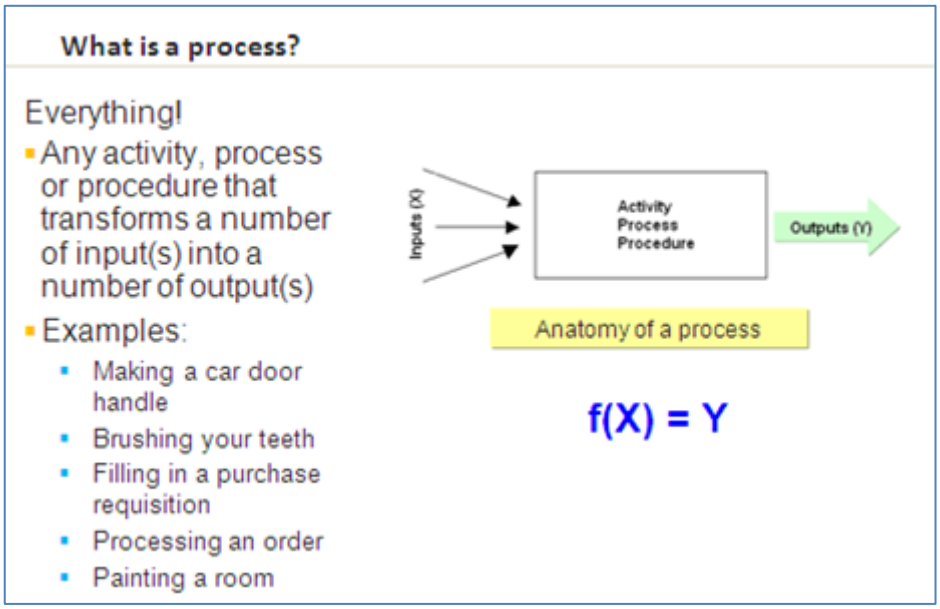
An activity or stream of activities that transforms or shapes raw materials or information to meet true customer needs

Inefficiencies

1. Waste	Overproduction, Motion, Transport, Waiting, Processing, Inventory, Defects, Space, Energy, Human Potential
2. Wasted Time	Variability is any deviation, in a service or product, that creates unnecessary costs
3. Inflexibility	Any inflexible practices that prevent you from meeting the customer's requirements at reasonable cost e.g. same shift patterns that do not meet customer needs

If the output doesn't match customer requirements the whole process is WASTE
Lean is about giving the customer what they want, not what the company thinks they want!

Let's get started on waste. In Lean a good definition of waste could be "doing anything that a customer does not need". This of course raises the question "just what is a customer?" To answer that we first need to look at processes.

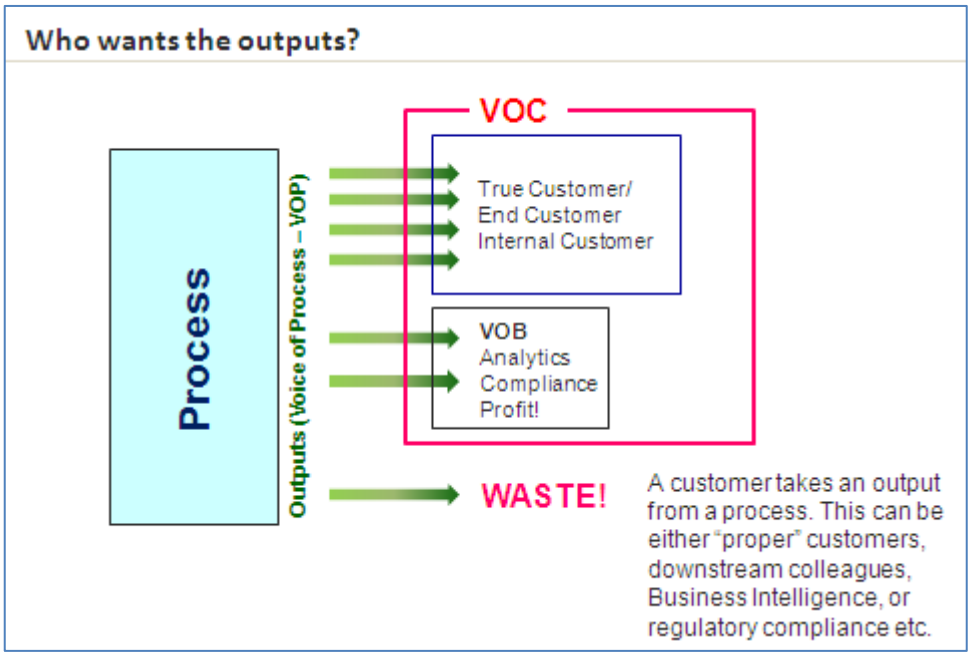


A process can be considered to consist of just three things:

1. Inputs – Stuff comes in, this could be iron & coal into a steel mill or a customer's details and their order into a mail order company. In L6S we often refer to inputs as "X"s.
2. Outputs – The process takes the inputs and transforms them into something else, including stuff that the customers want. So in the steel mill outputs are steel, hot air and slag whilst in the mail order we have the delivered package and waste paper (the picking lists etc.,). In L6S we often refer to the outputs of a process as "Y"s.
3. The last part is really the one in the middle: the transformation itself (this is also known as an X). The work or procedures that change what comes in to what goes out.

The relationship between X's & Y's is shown by the formula f(X) =Y which simply states that variance in your inputs and in the way you transform the inputs directly relates to variance in the outputs. So for example if you use a different coal with more sulphur in it your grade of steel will change proportionally, if each packer has their own routine then the time take to dispatch an order will also vary proportionately.

Now we have processes sorted let's get back to Customers!



A good definition of Customers is "those who consume outputs from a process". A customer will have definite views on what makes an output acceptable (hardness of the steel or the delivery time for example). We can split customers up into two major groups VOC and VOB (be careful as there are many interpretations of these terms).

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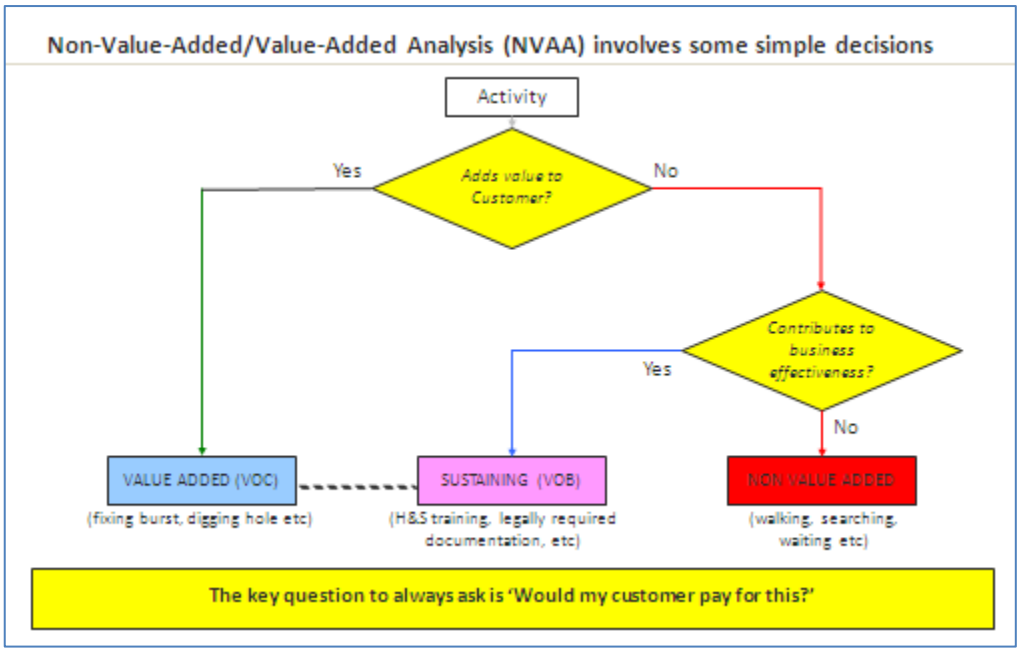
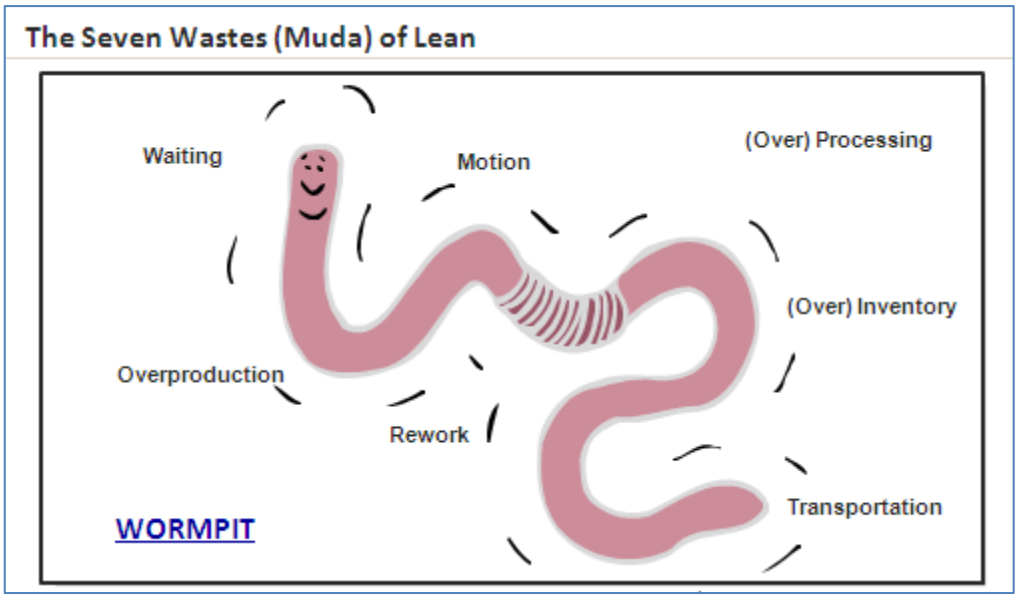
VOC stands for "Voice of the Customer" and depending on the discussion means either consumers (those that give the process a reason for being) or ALL groups who take outputs.

Sometimes however it is convenient to identify a subset of Customers as the Voice of the Business (VOB). Business will consume outputs (reports, Business Intelligence etc.), but that should not be the reason for running the process. VOB is usually taken to include compliance auditing and regulations as well.

So waste is simply something that none of your customers want.



We have tried to illustrate Lean waste in the above diagram. Value Add is something the consumer would be willing to pay for (an acceptable service or goods). Sustaining is something a consumer probably isn't interested in but still needs to happen. Anything else is waste and can be discarded without lowering the quality of the products in the Customers eyes.



Lean is BIG on waste. The above slide illustrates the 7 wastes of Lean. These are areas in a process that can be removed or diminished without lowering the quality of the outputs.

One way of looking at Lean is that it is a collection of tools & techniques to identify and remove waste from a process.

Here is a quick rundown of what each of the wastes describe:

Waiting

People or things waiting around without doing something of use (in the customer's eyes). Examples could be delays in getting authorisation for a necessary part, waiting for a purchase req to be authorised (especially for trivial amounts), a lorry waiting for spares to fix it, the time spent searching for something you have lost or your PC starting in the morning etc.

Overproduction

Making or doing things that are not required to meet the immediate (short term) demand. Once built a product has to be stored and often has a shelf life. You are gaining no payment while this is going on, however you will have to pay your suppliers.

Rework (often called "Defects")

Just the time and effort wasted when the "things" that you make are wrong or faulty – typo's in a report, wrong price in a quotation, a system incorrectly installed, a faulty widget. The Kaizen focus is to be always "right first time".

Motion/ Movement

The physical movement of people. Having to trek to the store for parts or pop downstairs to talk to co-workers is all waste.

Over Processing

Including features or doing extra tasks that the customer has no requirement for. Examples: filling in more fields on a form than are actually needed, purchasing re-usable syringes when they are always discarded after use, over packaging a product, producing a domestic alarm clock accurate to 1/100,000th of a second.

Over Inventory

Having too many of the wrong thing in stock (and not enough of the right things). Having too much work in progress or unfinished goods. For services this could be too many open or partially completed cases. Over production tends to lead to over inventory.

Transportation

Movement of physical items and data rather than the movement of people. For example sending forms that arrive in a local office to Head Office for processing which are then sent back to the local office for approval.

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Overview of 6 Sigma

6 Sigma came from Motorola. The "big idea" behind 6 Sigma is that variation in the outputs can only have **two** causes:

1. Variation in the inputs
2. Variation in how the inputs are processed

From this it follows that if we reduce the variation in the above we reduce variation in the outputs. Customers will only accept so much variance before they reject what we give them (and go elsewhere). Motorola considers that anything created from a process (what we have been calling outputs) is a "defect", So we can reduce defects in our process by reducing variation in what comes into it and how it is worked on.

Aims of 6 Sigma

- To improve processes by
  - Reducing defects
  - Reducing variation
- Many (but not all) of the tools are statistical in nature and 6 Sigma emphasises taking action based on fact rather than opinion or common belief
- The methodology uses a five phase model (DMAIC)

  Define
  Measure
  Analyse
  Improve
  Control

Common tools are:

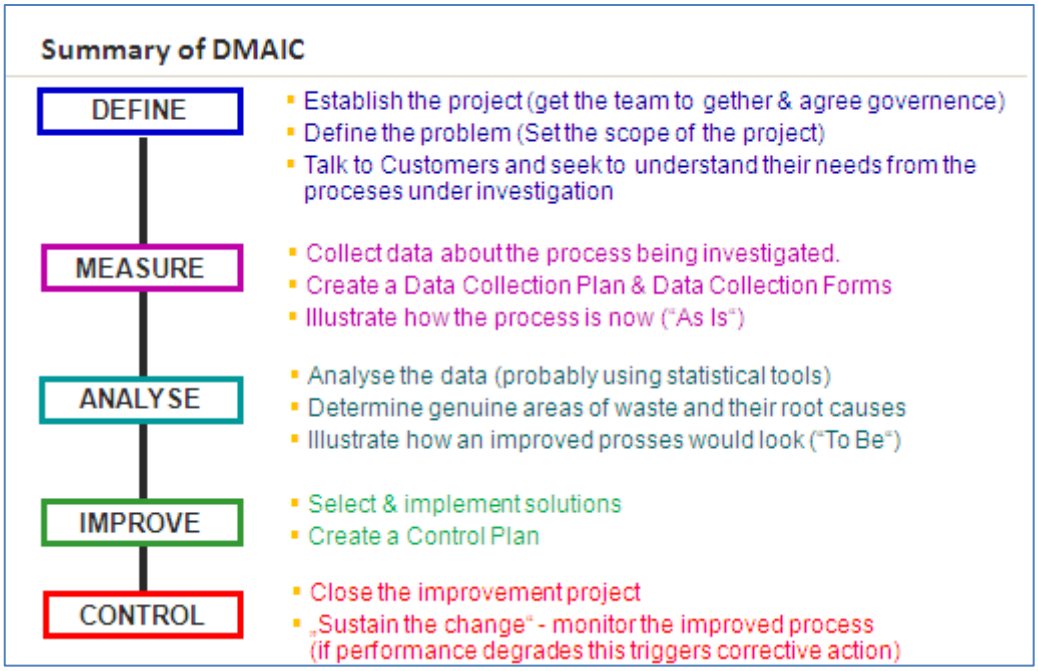
- SPC (Statistical Process Control)
- FMEA (Failure Mode and Effects Analysis) – a way of handling Risk

Motorola gave us other valuable tools such as SPC (a way of looking at processes statistically), the concept of Belts to denote competencies and of course the process improvement methodology DMAIC (pronounced Dom-ay-ik).

What is DMAIC?

- A five step methodology that provides a structured framework to improve a process
- The DMAIC framework helps ensure that improvement projects are:
  - Clearly defined
  - Clearly implemented
  - The results embedded in standard operations

As the above slides show, DMAIC is a good framework for identifying then removing some of the waste from a process. Why DMAIC? It is an acronym, each letter stands for a step in the process.



Think of each phase as a checklist of things that should be done at that point in time to achieve the improvement.
The main things to do each phase are as follows:

Define

This phase is concerned with setting up the improvement project and understanding what needs to be done (the scope). Things to do include: forming the team, identifying which part of a process the team will seek to improve, identifying what the customer desires from this process, and creating the project documentation.

Measure

In this phase the team investigates how the process is working now ("as is"), the team looks for areas of waste, inefficiencies and high defect so that they can develop a picture of where the process could be improved. Things to do include gathering data for which a **data collection plan** and **data collection forms** will be needed.

Analyse

The data gathered in the Measure phase is examined to identify areas which it is practical to improve and what the root causes of the wastes are.

Improve

In this phase the team decide on the best way to remove or mitigate a root cause (there are usually various ways to solve a problem); after this the team implements the selected solution. Things to do are implementation activities (to attack the root causes), piloting, training and documentation.

Control

This phase has two main aims:

1. Wrap up the project started in the Define phase
2. Sustain the change

Green Belt DMAIC projects are usually simple and informal, so wrapping up is basically throwing away your project paperwork, working out the potential savings and debriefing the team. Sustaining is making sure that the changes introduced in the Improve phase "stick" and putting an alerting mechanism in place to trigger action if the process slips back at a later date.

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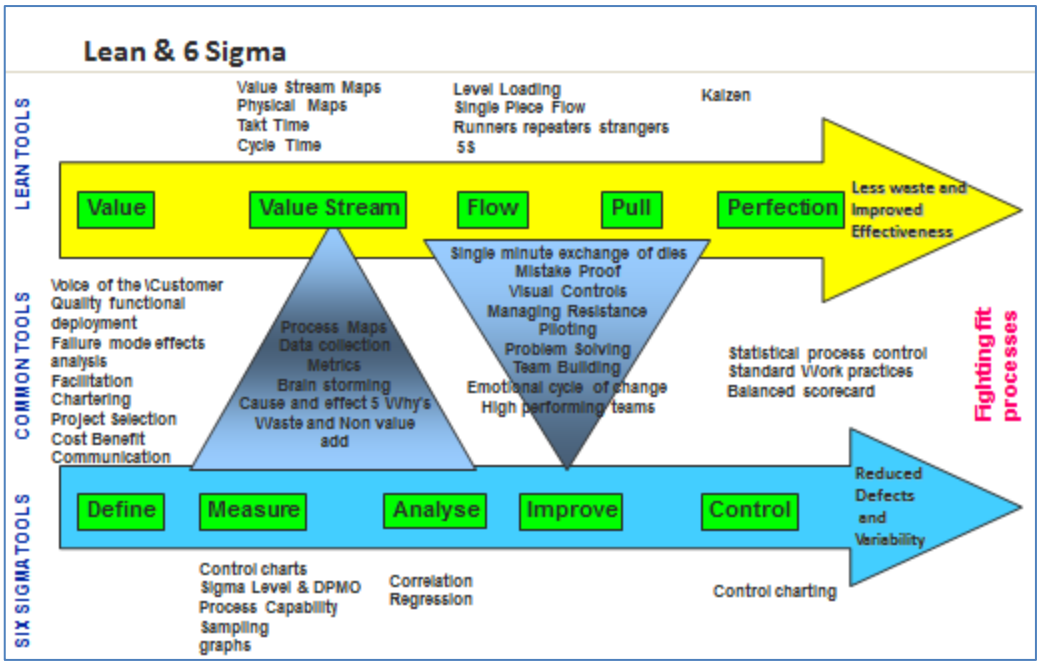
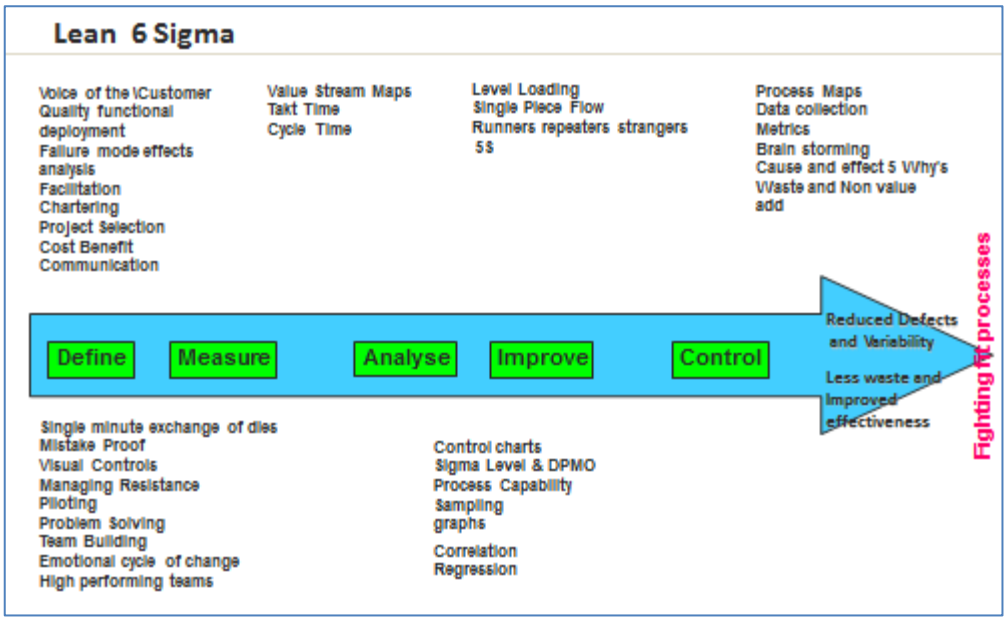
Lean 6 Sigma

It became obvious that Lean & 6 Sigma had huge synergies and worked best when combined. Lean 6 Sigma is that fusion .

TKA follow common practice of adding the Lean Philosophy and toolkit onto the 6 Sigma statistical tools and backbone of DMAIC giving us something rather like the following:





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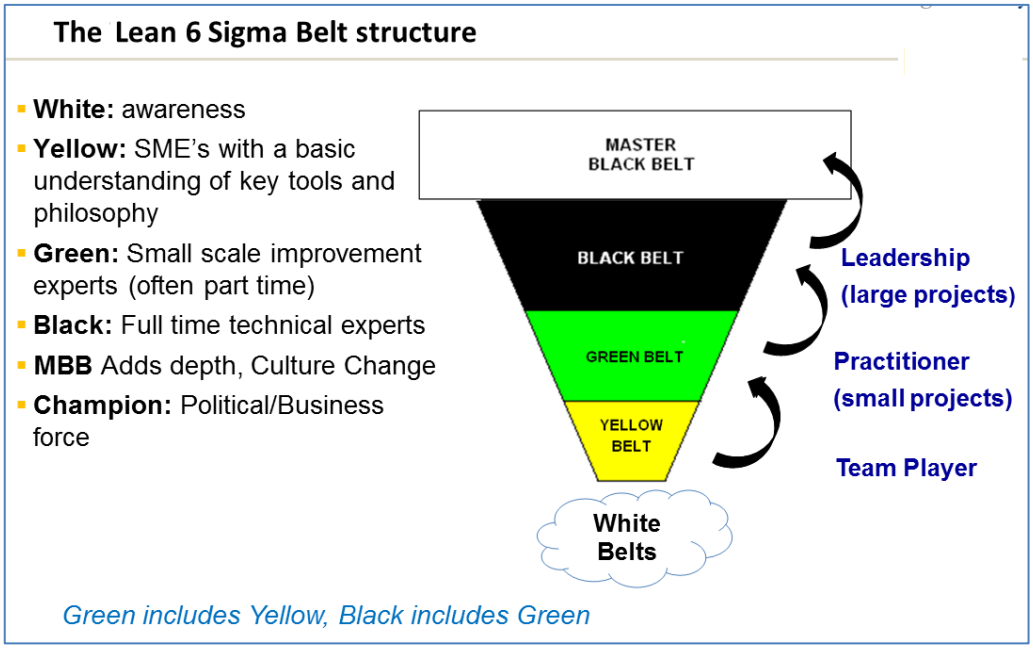
Lean 6 Sigma belt structure

The belting system is inherited from 6 sigma and is used to denote a person's capabilities in the subject. Once again there is a wide difference of opinion about the skills needed to perform a particular role so we have had to make an arbiter decision based on common practice.



Our definition of the five levels of belt is as follows:

White Belt

A basic awareness of what L6S is about and why an organisation would wish to embrace it (this study guide gets you to White Belt by the way!).

Yellow Belt

Yellow Belts have a very basic knowledge of DMAIC plus they can use many simple tools. These are the true SMEs (Subject Matter Experts), they are staff who work in a process and probably know best how to improve it. TKA considers that Yellow Belts will probably be managed by Green Belts.

Green Belt

Part-time practitioners (managers who add the GB role into their other duties). They will encourage and direct their Yellow Belts who will probably be staff directly managed by them. Green Belts are the technical backbone of the organisation and are capable of running small informal DMAIC projects.

Black Belt

Often full time practitioners (used to be always but this is changing as more and more senior people embrace L6S). TKA view this as a consultancy and support role (either internal or external). Black Belts train, motivate and support both Champions and practitioners of the art.

Master Black Belts

Black Belts who have more experience and who are comfortable working with the highest levels of management. Master Black Belts have a good working knowledge of associated Lean Techniques and methodologies such as Formal Kaizen and know how to introduce Lean Thinking into organisations in a sustainable and cost-effective way.

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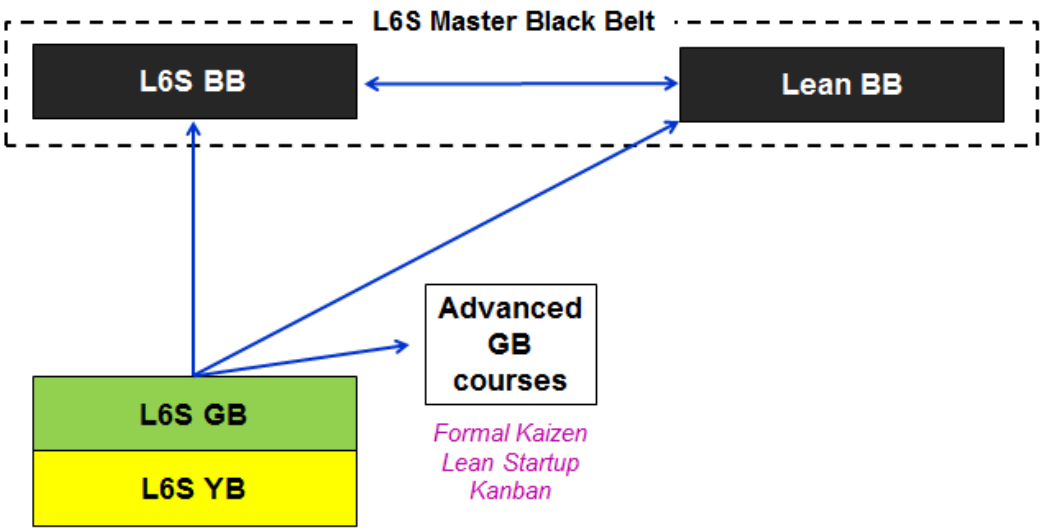
Course flow

The duration is of two days although for refreshers we sometimes do it in a single day

Topics are as follows:

- Introduction & Housekeeping
- Introduction to Lean 6 Sigma
- The DMAIC cycle
- Yellow Belt tools & techniques
    - The Lean Triad (5s, Poka Yoke, SMED)
    - Voice of the Customer
    - SIPOC
    - Value Stream Analysis
    - Heijunka
    - Variation & SPC
    - Defects & Process Capability
    - Identifying the Root Cause
    - Solving Problems
    - Kaizen Events
    - Lean Teams
- Yellow Belt Exam

Training Roadmap



Lean 6 Sigma Yellow Belt is the start of your journey and gives you the basic techniques and understanding to work effectively in a Lean 6 Sigma environment. With YB you will be capable of improving your own job or task and working with other Yellow Belts under the direction of Green Belts or Black Belts.

Lean 6 Sigma Green Belt builds on this to get you up to practitioner level (someone who can co-ordinate small teams and design and manage simple improvement initiatives).If you wish to pursue your journey after Green Belt then you have many options:

- You can add an additional skill such as Kanban or Formal Kaizen
- You can graduate to either Lean 6 Sigma Black Belt (which includes further statistical techniques) or Lean Black Belt (which includes the Advanced GB topics, Culture Change, and additional studies)
- If you elect to take both Lean 6 Sigma Black Belt AND Lean Black Belt then you will be certified as a Master Black Belt

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Course Scenario

The following serves as a background to the course discussions and exercises so please make sure you are familiar with it

Consolidated Sugar & Fats (CS&F) is a manufacturer and supplier of bagged confectionary. The company has three major revenue streams:

1. **Uncle Jim's**: A high quality nostalgic product range. The company produces a number of "old fashioned" sweets which it bags then sells to retail concerns world-wide. The company has established the brand as a luxury confectionary and uses only natural flavourings and ingredients in the recipe. The company also sells this product range direct to the public via their web site. The firm was founded in the early decades of the last century as a family run business and this brand was the original offering.

2. **Dibbers Supermarkets**: Like many supermarkets Dibbers offers its customers a pick & mix range. Dibbers are one of the top three UK supermarket chains and have an uncompromising attitude to keen pricing. CS&F has the contract to provide Dibbers pick & mix range. The contract specifies that CS&F must provide at least 12 varieties of sweet for a specified price per Kilo weight. The company have developed a specific line of sweets for Dibbers consisting of about 14 varieties of sweet at any time. The recipes and types are varied to meet the price target agreed. The supply contract has severe penalty clauses for late delivery of product. If the company fail to meet their production targets with Dibbers **twice** in a year the penalties will wipe out the whole years profit.

3. **Target Sweets**: The Target Sweet range is a value range of "Wholesale Only" bagged sweets which Uncle Jim's sells in bulk to distributors (wholesalers). Target basically duplicates the premium range but uses simpler and different packaging and synthesised flavours. Low cost is the main business driver for this brand. Uncle Jim's goes to great lengths to separate the two brands. For example the brand receives no mention on the corporate web site. The sweets are sold "informally" to a few UK wholesalers as and when a consignment is ready. To save money the packaging and contents have only been tested for compliance with UK regulations.

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About the company

Uncle Jim's was founded in the 1920s. The company has a factory near the small town of Minton in Somerset having established itself there in the 1930s. The company was established as a purveyor of quality confectionery and specialised in "boiled sweets" (sweets primarily composed of melted sugar and a small amount of flavourings). Output was always modest: initially the sweets were all handmade which severely limited production. In the early 1990s the last family member running the company retired and sold the business to a consortium of senior managers. A limited company was formed - Consolidated Sugar & Fats (CS&F) although the trading name continued to be "Uncle Jim's" as did the brand. Sweet production was automated with a single production line for all the boiled sweets which were produced in batches. Production and profits increased and then held stead till the present time. Even though CS&F is a public company they still strive to maintain the family firm ethos of the original founders, partly because of this Uncle Jim's sweets developed a good (if restricted) position in the global premium confectionary market.

The company started a new chapter in its history 18 months ago with the arrival of a new Managing Director Alan Forden. Alan is a keen supporter of Lean thinking and has instigated a Lean 6 Sigma culture into the company. He secured the Dibbers contract which necessitated an additional sweet production line (the Company were fortunate in being able to obtain a similar production line to their existing system when a rival closed).

The introduction of the second production line had several effects. Firstly it meant an increase in production staff. This was good news for the people of Minton as many manufacturing plants had closed over the previous few years. CS&F is now the largest employer in the town a matter that has not gone unnoticed. The Town Council has been particularly vocal in their appreciation: the Mayor herself has talked about a "Civic Partnership" with Uncle Jim's and has stated on several occasions that the Town Council considers itself a stakeholder in the company. Secondly with the introduction of the second line, it was discovered that there was now an excess of capacity. As a response to this, the Target value brand was introduced to absorb the waste. Target sweets are produced when either of the two production lines are not needed to manufacture any of the other brands (Uncle Jim's or Dibbers). The company is satisfied if the brand covers costs and perhaps makes a small profit. To achieve this Target sweets need to be sold (and quickly), with this in mind the company have deliberately set out to create a low cost sweet. Everything about Target is focused on cost effectiveness. For example the company did not invest time and money in creating new recipes but uses those in the Uncle Jim's range substituting cheaper synthetic flavourings for natural ones. The range is sold to distributors on an "as available" basis (the company refuses contractual supply agreements on the Target brand). Because of the low prices they never have any problem moving the output. Purchasing has strongly supported the Lean 6 Sigma initiative especially about the need to keep costs on the Target brand as low as possible. In fact about 30% of purchasing time is spent chasing down the lowest price for Target flavourings (both Sweet Brands use the same sugar)

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A short overview of the sweet manufacturing process

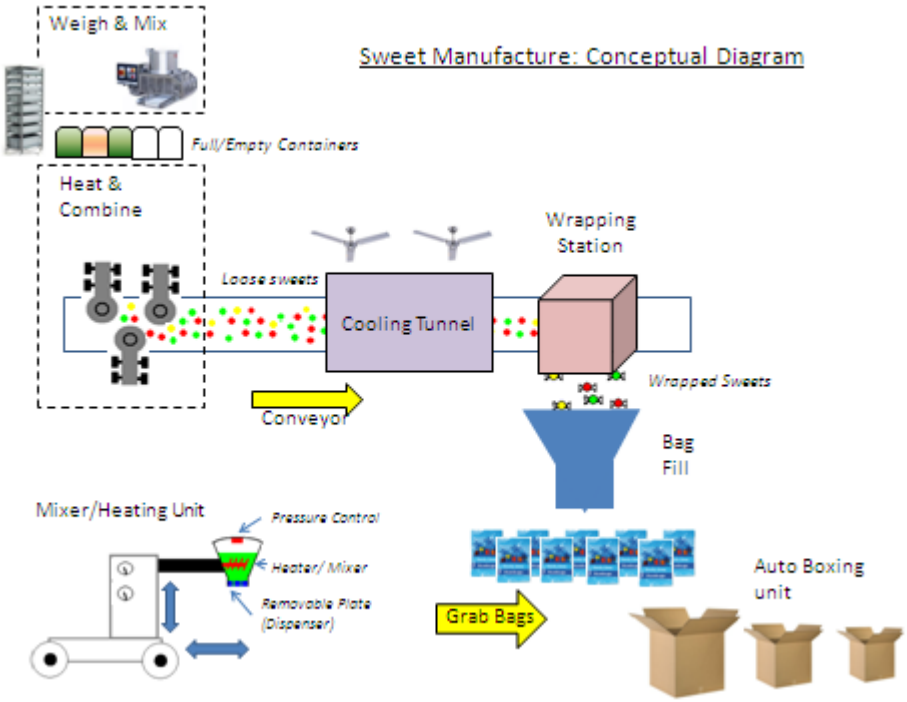
Note: *This description refers to the manufacture of boiled sweets; the company has additional production lines for chocolate based sweets and candy however these areas are not considered in this case study. Obviously the process has been simplified and shortened as the course is not about teaching you to make sweets!*

Each boiled sweet consists mainly of sugar with a small percentage of flavourings added.

The sugar and the flavourings are melted in vats and the contents stirred. An automatic nozzle at the base of each vat opens for a timed period to allow a drop of the mixture to fall onto the rolling belt underneath. This is quite a slow process as it depends on the viscosity of the mixture so several vats are needed to sustain the flow. Each vat has an aperture control that can be raised or lowered: these adjustments set the size of the sweet and influence its shape. The mixture cools during its fall but is still soft enough to flatten on impact with the belt. The production line has several vats so it is possible to create several different types of sweet at once; however the output of all the hoppers falls onto the same belt with the result that the sweets get mixed up. In practice this means that the production line is limited to **one** brand at a time however it is possible to produce several flavours at once if the intention is to bag the output as a mixture. The current practice is that if a mixture is to be bagged (e.g. Uncle Jim's Fruit Surprise) then the vats will hold different flavours, if only one type of sweet is needed (e.g. Uncle Jim's Old Fashioned Humbugs) then that mixture is spread across as many vats as required.

Once deposited, the sweets pass through a cooling tunnel which uses fans to force air across them. After this the sweets pass to the Wrapping Station which consists of several automatic wrapping devices. Each device has a sweet hopper and uses rolls of plastic film. A section of the film is cut, a sweet is fed from the hopper, then the film section wrapped round it. The wrapped sweets are deposited into open containers which are manually moved to the Bagging Station. At the Bagging Station the sweets are transferred into a hopper which has a funnel at the bottom with an automatic dispensing valve. Opened bags travel under the valve which feeds 100 sweets into it. The bags then travel to a sealing station which closes the bags and deposits them into cartons of 500.

The factory has two production lines handling boiled sweets; this means that the company can produce two types of sweet at a time. To manufacture another type of sweet the factory needs to stop a production line and thoroughly clean it to remove any trace of the previous ingredients - this takes about 2 hours so company policy is to manufacture in large batches (typically they produce enough of one sweet to satisfy 3 months demand)



Sales Cycles

Uncle Jim's range is very steady however there are peaks at Christmas and Mother's day and a seasonal slump in August for some reason. There is a high margin of profits on the brand and it has been found in the past that promotional marketing results in a heightened (but temporary) increase.

Dibbers has proved steady. The profit for this stream is not large however the level demand makes the business worthwhile to Uncle Jim's.

Target is very price sensitive, reducing the cost of a packet to the wholesaler can increase sales significantly, and unfortunately the profit margin on this brand is very low. CS&F only manufacture this brand when they have spare manufacturing capacity

All sweets are wrapped, Dibbers & Target use standard patterned wrappers, UJ brand is printed with logo the cost is the same. The plastic wrapping material is purchased from a specialist supplier. UJ uses a custom print however having paid a design fee years ago the supplier does not charge anything extra.

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